

Frequency Distribution (Quantitative)

C-I	f	CB	CF	Rf	$\times$	%f
1-3	17	0.5-3.5	17	$17/50$	2	$17/50 \times 100$
4-6	12	3.5-6.5	29	$12/50$	5	
7-9	6	6.5-9.5	35	$6/50$	8	
10-12	9	9.5-12.5	44	$9/50$	11	
13-15	6	12.5-15.5	50	$6/50$	14	

CB = Lower Limit - Upper Limit

MidPoint (X) = Average of Class Intervals.

CW = Diff. b/w class Intervals.

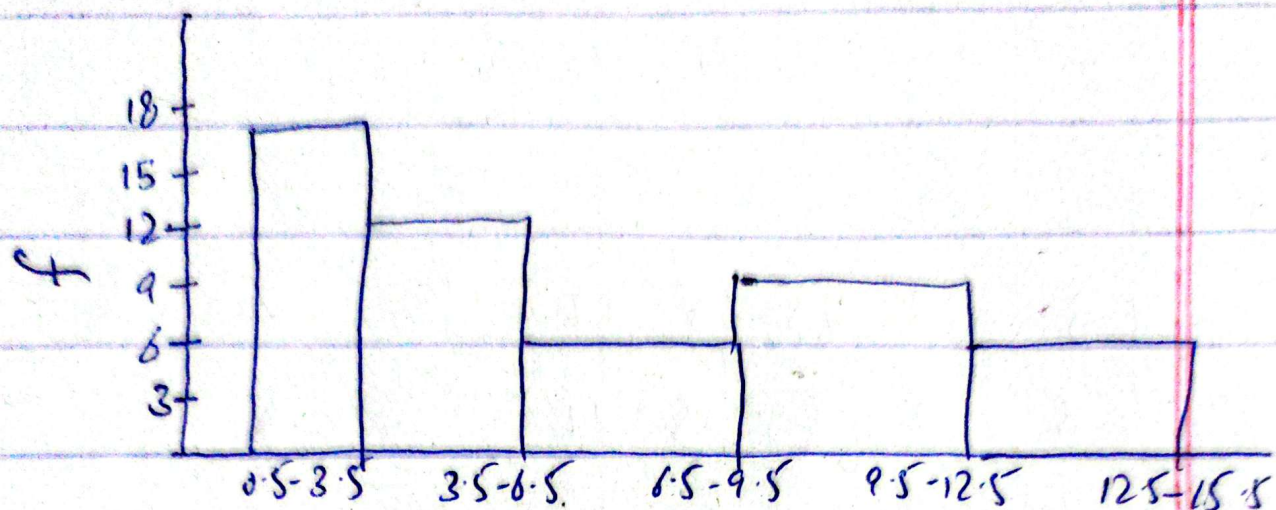
Histogram:

It is a graph of frequency distribution (Quantitative). Constructed by using f and CB.

- ⇒ To display frequency dist
- ⇒ To define shape of data

Types:

- (i) Symmetrical (Normal)
- (ii) Skew (Positive / Negative)



C-B

Frequency Polygon:

Constructed by f and x .

Measure of Central Tendency:

1- Mean: Average of data

$$\bar{X} = \frac{\sum X}{n}$$

Ex:

Mean of 12 values is 25. After revising the values we find out that one value is missing. By adding that value the new mean become 30. What is the missing value?

$$25 = \frac{\sum X}{12}$$

$$300 = \sum X \longrightarrow \textcircled{1}$$

$$30 = \frac{\sum X + A}{n+1}$$

$$30 = \frac{\sum X + A}{13}$$

$$390 = \Sigma X + A$$

Put ①

$$390 = 300 + A$$

$$\boxed{A = 90}$$

Types :

- Weighted Mean
- Combined Mean

Weighted Mean: When each value have different Preference.
Mean of multiple means.

$$\bar{X}_w = \frac{\Sigma W_i X_i}{\Sigma W_i} = \frac{W_1 X_1 + W_2 X_2 + \dots}{W_1 + W_2 + \dots}$$

W = weights

X = data value

Combined Mean: Mean of multiple means.

$$\bar{X}_c = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + \dots}{n_1 + n_2 + \dots}$$

Ex: A software "A" detect 13 outliers when we execute 20 times and software "B" detects 19 outliers when we executed 30 times.

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$= \frac{20(13) + 30(19)}{20 + 30}$$

$$= \frac{260 + 570}{50}$$

$$= 16.6$$

$$\cong 16$$

2- Median:

Middle value of the data after arranging the data in ascending or descending order. which divides data in two equal parts.

- 50% observation fall above the median.
- 50% observation fall below the median.

It is not affected by outliers, so it is superior than mean.

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value of data.}$$

Properties:

- Quantiles:

1) Quantiles: It divides dataset into 4 equal parts. Q_1 (Lower Quartile) contain 25% data below and 75% above. Q_3 (Upper Quartile) contain 75% data below and 25% above.

2) Deciles: Deciles are the 9 values which divides the data in 10 equal parts. There is 10% data between any 2 points.

3) Percentiles: It divides data into 100 equal parts by 99 values.

1) Quartile.

$$Q1 = \frac{1(n+1)^{th}}{4}$$

$$Q2 = \frac{2(n+1)^{th}}{4}$$

$$Q3 = \frac{3(n+1)^{th}}{4}$$

3- Mode:

Mode is the most frequent value in the data. It is used as a average for qualitative data as well as for quantitative data.

Median:

3.91

2.95, 3.15, 3.75, 4.32, 5.51, 5.91,
6.51, 6.71, 7.01, 7.11, 7.18

$$= \frac{5.51 + 5.91}{2}$$

$$\frac{121}{2}$$

6.5

$$= \frac{11.42}{2}$$

$$M, Q_2 = \boxed{5.91}$$

$$Q_1 = \frac{1(12+1)}{4} = \frac{13}{4} = \boxed{3.25}$$

$$= 3.75 + 0.25(3.91 - 3.75)$$

$$Q_1 = \boxed{3.79}$$

$$Q_3 = \frac{3(12+1)}{4} = \frac{39}{4} = 9.75$$

$$= 6.71 + 0.75(7.01 - 6.71)$$

$$Q_3 = \boxed{6.93}$$

Measurement Scales:

Nominal $\longrightarrow$ Qualitative Data

Ordinal $\longrightarrow$ Ordered Data

Interval $\longrightarrow$ Continuous Data

Ratio

Define Shape of Data:

(i) Histogram

(ii) Mean and Median: If mean is equal median, it means that data is symmetrical.

If mean is greater than median, it is positively skewed.

If mean is smaller than median, it is negatively skewed.

Measure of Dispersion:

(ii) Variance/Standard Deviation

$$S^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$SD/S = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2}$$

★ Coefficient of variation is used when data have different units and we have to compare these datasets. It converts the data into percentages or proportions.

The data having large coefficient of variation considered as more superior or more variant. and data having less coefficient of variation means this data is consistent.

$$CV = \frac{SD}{\bar{x}} \times 100$$

(ii) IQR,

$$\text{Lower fence} = Q1 - 1.5(IQR)$$

$$IQR = Q3 - Q1$$

$$\text{Upper fence} = Q3 + 1.5(IQR)$$

EDA: It describes basic features of data.

(i) Min

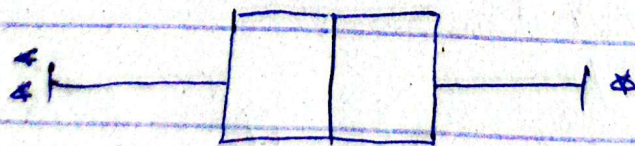
(ii) Q1

(iii) Median

(iv) Q3

(v) Max

BoxPlot:



If $Q1$ and $Q3$ are at same distance from $Q2$, the data is symmetrical.

If $Q3$ distance is greater than $Q2$, Positive Skewed.

If $Q1$ distance is greater than $Q3$, Negative Skewed.